

Ali Vaziri

Machine Learning, Scalable Simulations, Robotics, and Control
Ph.D. Student, Mechanical Engineering
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I. Professional Summary

AI/ML engineer and Ph.D. researcher with 5+ years of experience developing learning, planning, and control systems for autonomous vehicles, robotics, and energy systems. My work spans multimodal trajectory prediction, physics-guided learning, Bayesian inference, model predictive control, and distributed PyTorch training. At the General Motors Cruise team, I build large-scale data, training, and evaluation pipelines for simulation. Research implementations have achieved up to $200\times$ faster computation, with publications in *IEEE Transactions on Robotics*, *Applied Thermal Engineering*, IEEE CDC, and ACC.

II. Industry Research Experience

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| Summer 2026
Sunnyvale, CA | AI/ML Software Engineering Intern, General Motors – Cruise Team
Scalable Scenario Conversion Using Machine Learning <ul style="list-style-type: none">- Built an End-to-End multimodal ML system in scale that reconstructs missing vehicles' motion in recorded-driving simulations, supporting more complete and realistic testing of autonomous-driving behavior.- Built cloud data and featurization pipelines for more than 38,000 highway scenes from approximately 1 TB of raw tracking data, using Google Cloud Storage, Dataflow, and memory-mapped feature stores.- Scaled training across 8 H100 GPUs using PyTorch DistributedDataParallel, NCCL, and mixed-precision computation, cutting wall-clock training time from 12 hours to 3 hours in RoboFlow cloud workflows.- On a held-out evaluation of approximately 58,000 vehicle trajectories, reduced best-candidate average position error by 24% relative to company's baseline.- Reworked full-test evaluation around streaming data, parallel loading, and memory-mapped features, reducing runtime from more than 60 minutes to approximately 9 minutes and enabling faster model comparisons and data-quality studies.- Implemented scenario-level collision detection among surrounding vehicles and between the ego vehicle and surrounding vehicles. The metric checks oriented vehicle footprints at 10 Hz, removes duplicate collision pairs, and complements trajectory accuracy with collision, smoothness, and appearance-timing measurements. |
| Summer 2024
Cambridge, MA | Research Intern, Mitsubishi Electric Research Laboratories (MERL)
Machine Learning Modeling and Optimization <ul style="list-style-type: none">- Developed a physics-constrained Neural ODE-GRU model for vapor-compression HVAC systems, improving forecasting accuracy by 14% and runtime by $5.7\times$ relative to the previous modeling approach.- Combined continuous-time neural dynamics, recurrent sequence modeling, and known physical constraints to obtain accurate ML predictions from limited data.- Delivered reusable PyTorch software for data pipeline, model training, evaluation, and GPU-accelerated control experiments. |

III. Education

- Ph.D. Mechanical Engineering – Michigan State University 2026–present
Research focus: Scientific machine learning, Bayesian inference, and optimal control
- Ph.D. Mechanical Engineering – University of Kansas 2023–2025
Transferred to Michigan State University
- M.S. Mechanical Engineering – University of Kansas 2023–2025
GPA: 4.00/4.00 – Thesis: *Heavy-Tailed Bayesian Motion Planning for Autonomous Vehicles*
- B.S. Marine Engineering – Sharif University of Technology 2016–2021
GPA: 3.55/4.00 – Thesis: *Collision-Free Marine Waste-Collection Robot*

IV. Academic Research Experience

- 2023–present **Graduate Research Assistant, Michigan State University**
Scientific Machine Learning and Intelligent Control/Estimation
- Reformulated model predictive control (MPC) as Bayesian inference for neural state-space vehicle models; tested configurations solved up to **200× faster** than CasADi/IPOPT benchmarks.
 - Developed GPU-compatible matrix-variate ensemble Kalman smoothing for high-dimensional spatial models, achieving up to **18.6× faster computation** and approximately **11× lower memory use** than a vectorized GPU implementation.
 - Designed robust motion planning with Student's-*t* distributions, sequential Monte Carlo, and ensemble Kalman smoothing for non-Gaussian uncertainty and outliers.
 - Trained GRU and ResNet vehicle-dynamics models from real-world driving data and connected them to Bayesian planning and conventional trajectory optimization pipelines.
 - Continuous-time optimal control of Neural ODEs and high-dimensional physical systems, including Burgers and Navier–Stokes PDEs.

V. Technical Skills

Machine Learning and Modeling: Deep learning, generative modeling, multimodal learning, sequence and time-series modeling, convolutional neural networks (CNNs), recurrent neural networks, Neural ODEs, probabilistic modeling, physics-guided learning, scientific machine learning, system identification

Control, Planning, and Estimation: Model predictive control, Bayesian inference, trajectory optimization, robust motion planning, Kalman filtering and smoothing (EKF, UKF, EnKF/EnKS), particle filtering, sequential Monte Carlo, uncertainty-aware planning

Programming and ML Systems: Python, C/C++, CUDA, PyTorch, TensorFlow, NumPy, SciPy, Distributed-DataParallel (DDP), NCCL, mixed-precision training, distributed GPU computing, performance profiling, unit and integration testing

Autonomous-Vehicle Simulation and Evaluation: Road-to-Sim, Matrix, Hydra, Galileo, Webviz, scenario conversion, simulation testing, trajectory evaluation, collision and quality metrics

Robotics and Optimization Tools: ROS, MuJoCo, PyBullet, CasADi, IPOPT, MATLAB/Simulink, Solid-Works

Cloud, Data, and Development Tools: Google Cloud Storage, Dataflow, RoboFlow, memory-mapped datasets, Bazel, Linux, Git